PRADEEP LOGANATHAN

DATA SCIENCE STUDENT | PYTHON | SQL | MACHINE LEARNING

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PROFILE

B.Sc. Physics student focused on Data Science and Machine Learning, with hands-on experience solving real-world problems through data analysis and machine learning. Skilled in Python, SQL, EDA, feature engineering, model evaluation, explainability, and deployment. Built independent projects analyzing 210 Chennai bus stops, 47 drainage covers, and 6.3M+ financial transactions.

PROJECTS

Chennai Bus Stop Audit & Accessibility Analysis

Independent Data Analysis Project | Chennai

- Independently analyzed 210 Chennai bus stops across 12 analytical topics to investigate public-transport infrastructure, accessibility, and pedestrian conditions.
- Cleaned and explored an 84-column audit dataset, analyzing patterns across accessibility and infrastructure variables.
- Analyzed pedestrian and wheelchair accessibility at bus stops; 155 of 210 stops (73.8%) did not allow users to pass on the footpath without stepping onto the road.
- Performed cross-topic analysis to identify infrastructure and accessibility patterns and translate findings into practical public-transport improvement insights.

Data-Driven Analysis of Uneven Drainage Covers in Thiruvottiyur

Independent Community Infrastructure Research

- Independently designed and conducted a field survey covering 47 drainage covers across Thiruvottiyur.
- Collected and analyzed GPS location, cover condition, height difference, pedestrian traffic, vehicle traffic, risk level, priority, and resident feedback.
- Found 25 of 47 covers (53.2%) requiring maintenance, including 11 raised, 9 sunken, and 5 broken covers.
- Combined cover condition with pedestrian and vehicle traffic to identify 8 high-risk covers requiring immediate repair and prioritize maintenance resources.
- Proposed data-guided recommendations including periodic inspections, community reporting, and repair prioritization.

Real-Time Fraud Detection System

MachineLearning | XGBoost | SHAP | FastAPI | Streamlit | Docker

- Built an end-to-end fraud detection system using 6.3M+ financial transactions containing 8,213 fraud cases.
- Engineered transaction-risk features including balance differences, transaction anomalies, and amount-to-balance ratios.
- Trained and compared Logistic Regression, Random Forest, and XGBoost using ROC-AUC and Average Precision.
- Applied probability threshold optimization to prioritize fraud recall and reduce false-negative risk in an extremely imbalanced classification problem.
- Used SHAP to explain model predictions and identify influential fraud-risk features.
- Deployed the model through FastAPI and Streamlit, using Docker and Joblib in the deployment pipeline.

TECHNICAL SKILLS

Programming: Python, SQL
Data Analysis: Pandas, NumPy, Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Data Visualization
Machine Learning: Scikit-learn, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, XGBoost
Model Evaluation: Cross-Validation, ROC-AUC, Precision-Recall, Average Precision, Threshold Optimization, Confusion Matrix
Explainability: SHAP
Deployment: FastAPI, Streamlit, Docker, Joblib
Tools: Jupyter Notebook, Git, GitHub, Power BI, Excel

TRAINING & CERTIFICATIONS

Data Science Training — IntrnForte | Dec 2025
Data Science fundamentals, statistics, Python, data analysis, and Machine Learning
Machine Learning Certification — Simplilearn | Dec 2025
British Airways Data Science Job Simulation — Forage | May 2026
Tata Data Visualisation: Empowering Business with Effective Insights — Forage | Dec 2025
Data Labeling Job Simulation — Forage | Dec 2025

EDUCATION

B.Sc. Physics — The New College, Chennai | 2024–2027